

2002

**HIGHER SCHOOL CERTIFICATE
EXAMINATION**

Chemistry

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Draw diagrams using pencil
- Board-approved calculators may be used
- A data sheet and a Periodic Table are provided at the back of this paper
- Write your Centre Number and Student Number at the top of pages 9, 13, 17 and 21

Total marks – 100

Section I Pages 2–23

75 marks

This section has two parts, Part A and Part B

Part A – 15 marks

- Attempt Questions 1–15
- Allow about 30 minutes for this part

Part B – 60 marks

- Attempt Questions 16–27
- Allow about 1 hour and 45 minutes for this part

Section II Pages 25–30

25 marks

- Attempt ONE question from Questions 28–32
- Allow about 45 minutes for this section

Section I

75 marks

Part A – 15 marks

Attempt Questions 1–15

Allow about 30 minutes for this part

Use the multiple-choice answer sheet.

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
 A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

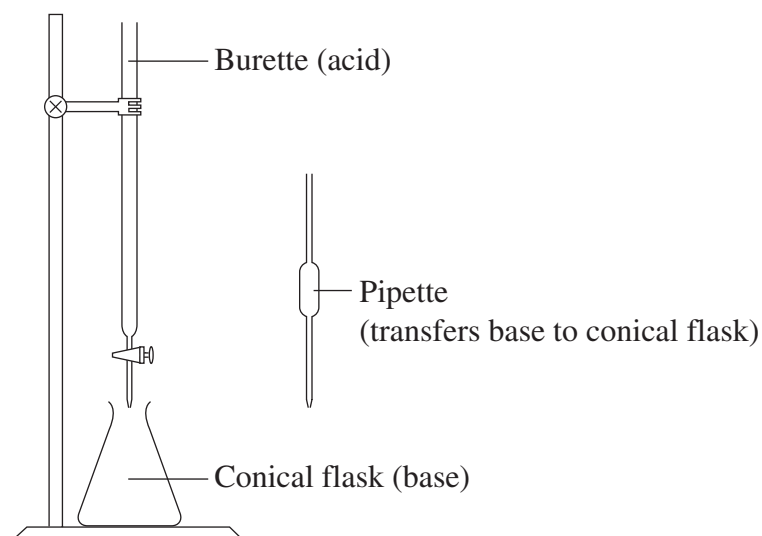
If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word *correct* and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐
 correct
 ↙

- 1 Which conditions would be best for the fermentation of sugars by yeast?
- (A) Low oxygen concentration and a temperature between 25°C and 35°C
(B) High oxygen concentration and a temperature between 25°C and 35°C
(C) Low oxygen concentration and a temperature between 45°C and 60°C
(D) High oxygen concentration and a temperature between 45°C and 60°C
- 2 What is the catalyst for the conversion of ethanol to ethene?
- (A) NaOH
(B) H₂SO₄
(C) HNO₃
(D) Pt
- 3 Which explanation best describes why an understanding of electrolysis has made a significant impact on the production of materials?
- (A) Electrolysis reactions do not require an energy source.
(B) Electrolysis can be used to split water into hydrogen and oxygen.
(C) Electrolysis provides a way to make non-spontaneous reactions occur.
(D) Electrolysis can be used to produce aluminium.
- 4 Which statement concerning galvanic cells is correct?
- (A) Oxidation occurs at the anode.
(B) They are also known as electrolytic cells.
(C) The cathode is assigned a negative charge.
(D) An external power source must be present.
- 5 What arrangement of an electrolytic cell would electroplate a silver coin with copper?

	<i>Cathode</i>	<i>Anode</i>	<i>Electrolyte</i>
(A)	copper	silver coin	copper sulfate
(B)	silver coin	copper	silver nitrate
(C)	copper	silver coin	silver nitrate
(D)	silver coin	copper	copper sulfate

- 6 Which is amphoteric?
- (A) H_2SO_4
 (B) NH_4^+
 (C) HCO_3^-
 (D) SO_4^{2-}
- 7 What did the Brønsted–Lowry definition of acids identify that made it a significant improvement over earlier definitions?
- (A) Acids contain hydrogen.
 (B) Acids are proton donors.
 (C) Acids contain oxygen.
 (D) Acids are electron-pair acceptors.
- 8 In a titration, an acid of known concentration is placed in a burette and reacted with a base that has been pipetted into a conical flask.



What should each piece of glassware be rinsed with immediately before the titration?

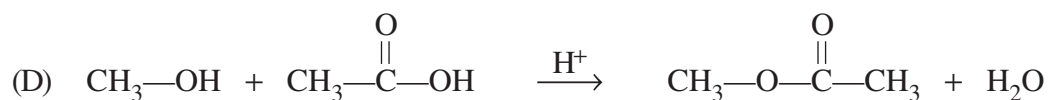
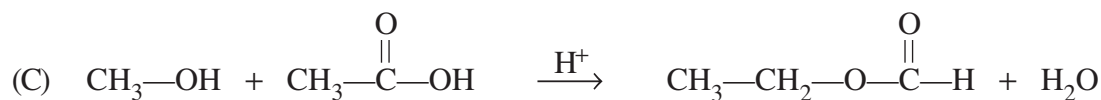
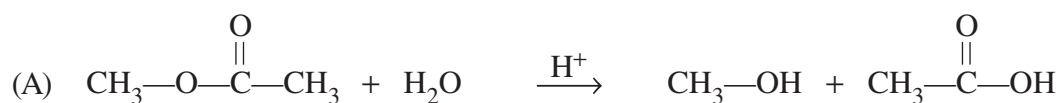
	<i>Burette</i>	<i>Pipette</i>	<i>Conical flask</i>
(A)	acid	base	water
(B)	water	water	water
(C)	acid	base	base
(D)	water	water	base

- 9 The following list of steps refers to an experimental plan for making an ester in a flask. Some of the steps in the list may NOT be required for this experiment. The steps are NOT in the correct sequence.

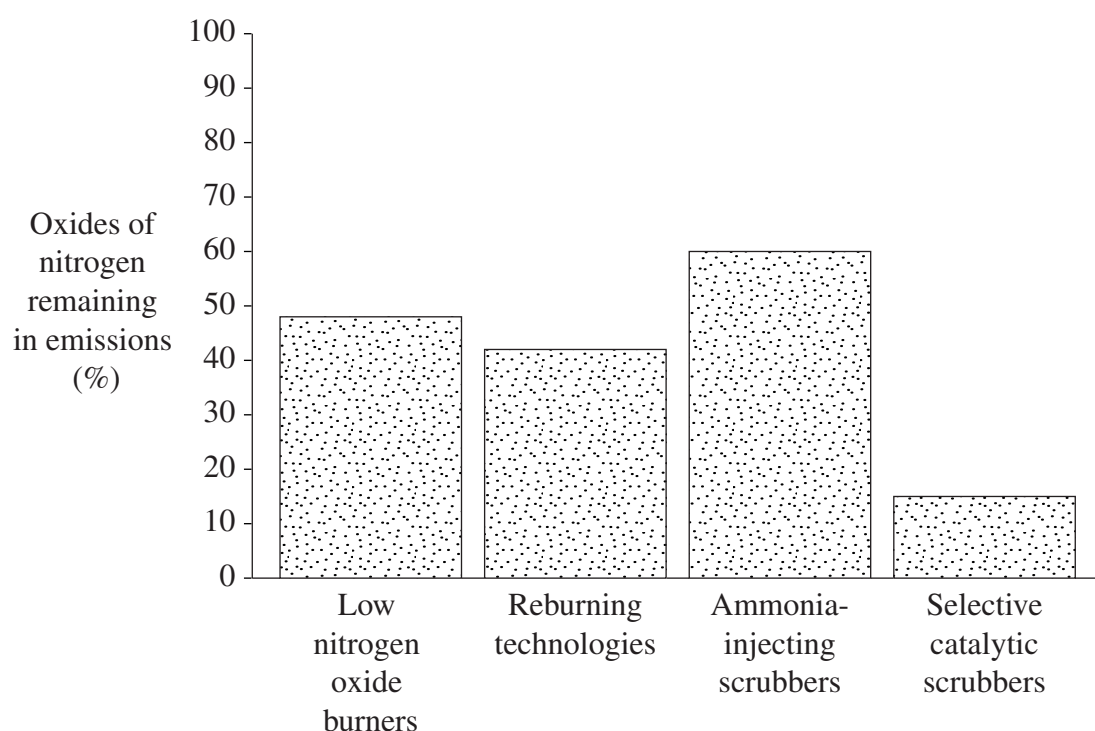
1. Heat the mixture under reflux.
2. Add three drops of concentrated sulfuric acid.
3. Add 1 mL of ethanol.
4. Add 1 mL of ethene.
5. Add 1 mL of ethanoic acid.
6. Distil the mixture.
7. Add three drops of phenolphthalein indicator.

Which alternative is the best sequence for making an ester?

- (A) 3, 5, 7, 1
 (B) 4, 3, 7, 6
 (C) 5, 4, 2, 6
 (D) 5, 3, 2, 1
- 10 Which equation represents esterification?



- 11 A car engine burns fuel with insufficient air. Which substance would be emitted in the exhaust in higher levels than from an engine with a correct fuel to air ratio?
- (A) Carbon dioxide
(B) Carbon monoxide
(C) Oxygen
(D) Water
- 12 Oxides of nitrogen are produced by the combustion of coal. The percentage of these compounds remaining in emissions can be reduced using the methods named in the graph.



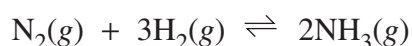
Which of the following is the most effective method for removing oxides of nitrogen produced by the combustion of coal?

- (A) Low nitrogen oxide burners
(B) Reburning technologies
(C) Ammonia-injecting scrubbers
(D) Selective catalytic scrubbers

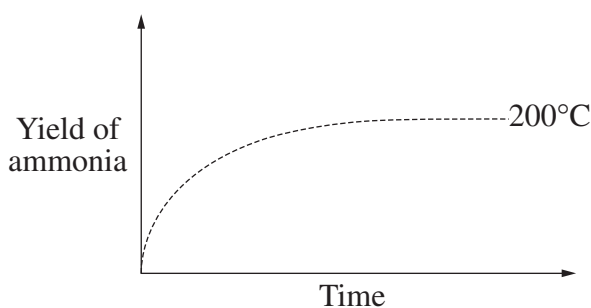
- 13 The Haber Process for producing ammonia was developed early in the twentieth century. What was the major advantage of its development?

(A) A government sold the process to other governments.
(B) The inventor sold the process for a great deal of money.
(C) It provided a source of nitrogen for farming and industry.
(D) It provided jobs for many who were unemployed.

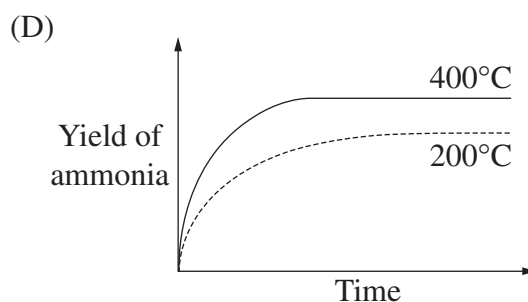
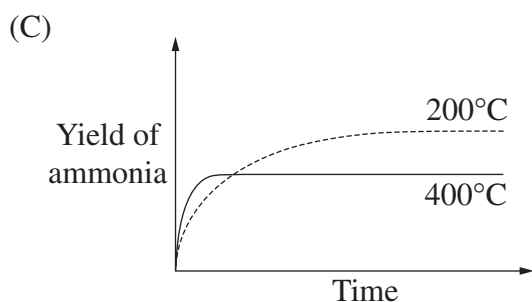
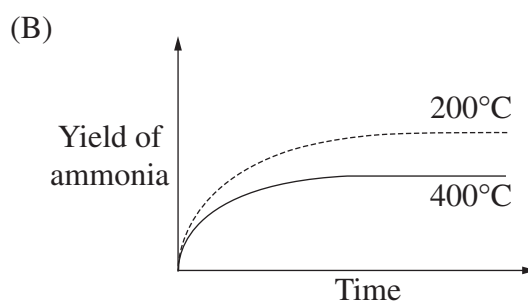
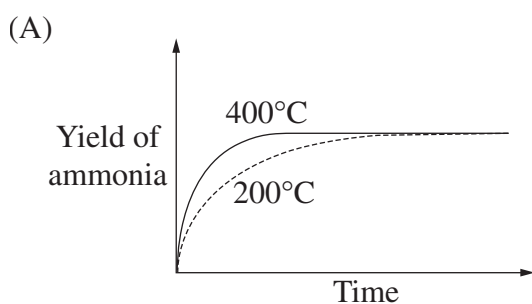
- 14 Ammonia is produced from hydrogen and nitrogen, according to the equation:



The graph shows the yield of ammonia produced at 200°C and 100 kPa.



Which graph shows a correct comparison of the yield of ammonia produced at a temperature of 400°C and 100 kPa with the yield produced at 200°C and 100 kPa?



- 15 The table gives the results of chemical tests for some cations and anions.

(ppt = precipitate)

<i>Ion</i>	<i>Add cold 0.1 M HCl</i>	<i>Add 0.1 M KSCN</i>	<i>Add 0.1 M Na₂CO₃</i>	<i>Add 0.1 M AgNO₃</i>
Ca ²⁺	no change	no change	white ppt	no change
Fe ³⁺	no change	red colour	brown ppt	no change
Ba ²⁺	no change	no change	white ppt	no change
Pb ²⁺	white ppt	no change	white ppt	no change
Cl ⁻	no change	no change	no change	white ppt

When a group of students performed the above tests on an unknown solution they obtained the following results:

<i>Add cold 0.1 M HCl</i>	<i>Add 0.1 M KSCN</i>	<i>Add 0.1 M Na₂CO₃</i>	<i>Add 0.1 M AgNO₃</i>
no change	no change	white ppt	white ppt

Which conclusion is consistent with these results?

- (A) The sample contained both CaCl₂ and BaCl₂.
- (B) The sample contained both CaCl₂ and PbCl₂.
- (C) The sample contained both FeCl₃ and PbCl₂.
- (D) The sample contained both FeCl₃ and BaCl₂.

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Centre Number

Section I (continued)

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Part B – 60 marks

Attempt Questions 16–27

Allow about 1 hour and 45 minutes for this part

Student Number

Answer the questions in the spaces provided.

Show all relevant working in questions involving calculations.

Marks

Question 16 (6 marks)

You have carried out a first-hand investigation to compare the reactivity of an alkene with its corresponding alkane.

- (a) State the name of the alkene. 1

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- (b) Outline a procedure to compare the reactivity of this alkene with its corresponding alkane. 2

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- (c) Describe the results obtained from this first-hand investigation and include relevant chemical equations. 3

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Question 17 (3 marks)

Explain why alkanes and their corresponding alkenes have similar physical properties, but very different chemical properties.

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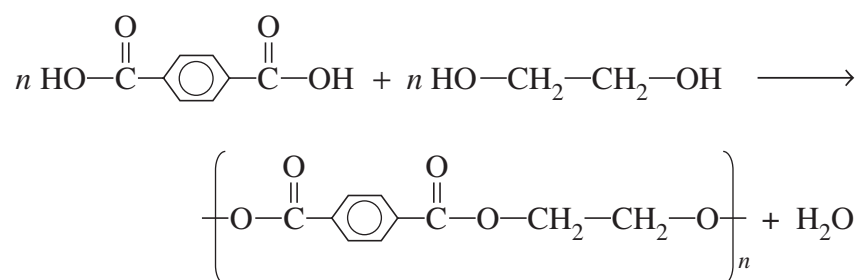
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- 1**



- 5

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Centre Number

Section I – Part B (continued)

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Student Number

Marks

Question 19 (5 marks)

- (a) Describe the conditions under which a nucleus is unstable.

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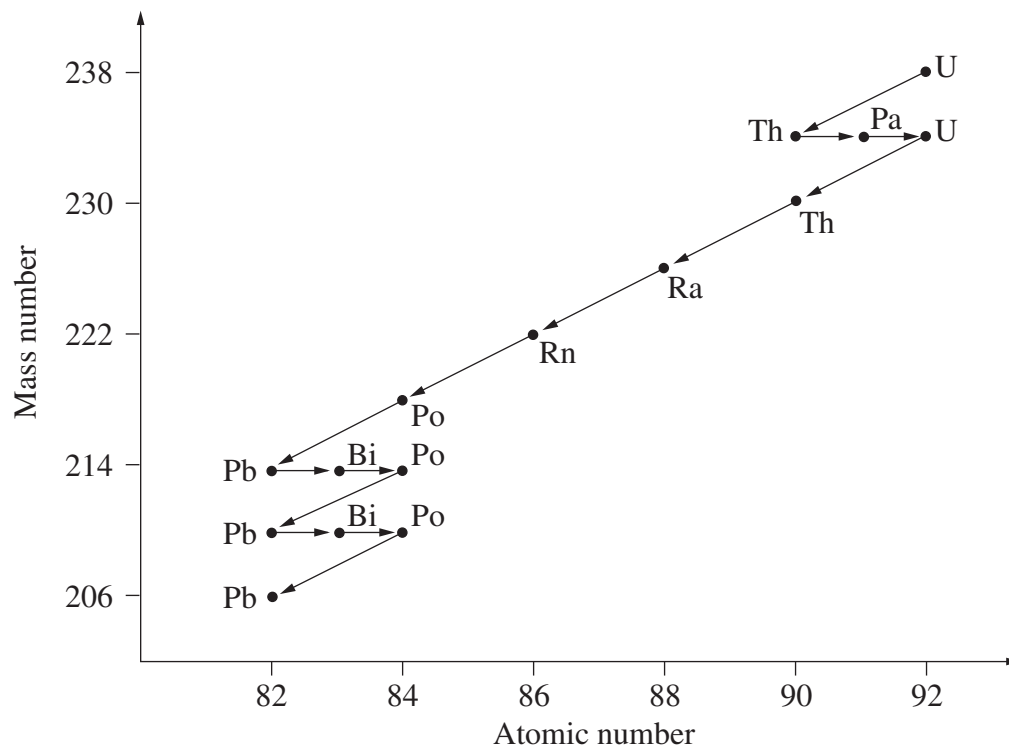
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Question 19 continues on page 14

Question 19 (continued)

- (b) The following is a flow diagram showing the sequence of products released during the decay of uranium.

3



Use examples from the flow diagram to describe processes by which an unstable isotope undergoes radioactive decay.

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End of Question 19

Question 20 (4 marks)

- (a) Identify ONE common household base.

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- (b) A student used indicators to determine whether three colourless solutions were acidic or basic. The indicators used are shown in the table.

3

<i>Indicator</i>	<i>Colour change</i>	<i>pH range</i>
Methyl orange	red to yellow	3.2–4.4
Methyl red	red to yellow	4.8–6.0
Thymol blue	yellow to blue	8.0–9.6
Alizarin	red to purple	11.0–12.4

Samples of each solution were tested with the indicators. The colours of the resulting solutions are shown in the table.

<i>Indicator added</i>	<i>Colour of solution A</i>	<i>Colour of solution B</i>	<i>Colour of solution C</i>
Methyl orange	yellow	yellow	yellow
Methyl red	yellow	yellow	yellow
Thymol blue	blue	blue	yellow
Alizarin	purple	red	red

The student concluded that each of the three solutions tested was basic. Assess the validity of this conclusion.

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[illegible]

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Centre Number

Section I – Part B (continued)

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Student Number

Marks

Question 22 (5 marks)

Solutions of hydrochloric acid, acetic acid and sulfuric acid were prepared. Each of the solutions had the same concentration (0.01 mol L^{-1}). The pH of the acetic acid solution was 3.4.

0.01 mol L^{-1} hydrochloric acid	0.01 mol L^{-1} acetic acid pH = 3.4	0.01 mol L^{-1} sulfuric acid
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- (a) Calculate the pH of the hydrochloric acid solution. 1

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- (b) Compare the pH of the sulfuric acid solution to the pH of the hydrochloric acid solution. Justify your answer. (No calculations are necessary.) 2

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- (c) Explain why the acetic acid solution has a higher pH than the hydrochloric acid solution. 2

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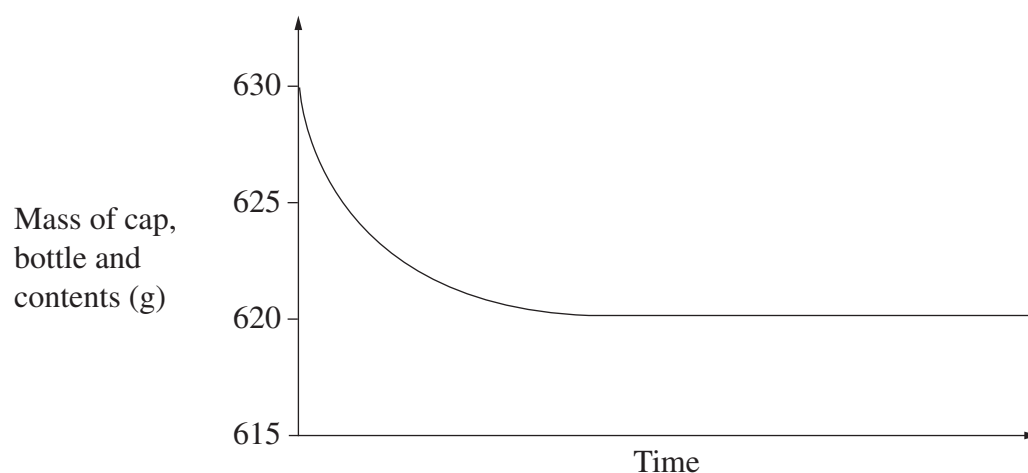
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Question 23 (4 marks)

A bottle of soft drink was placed on an electronic balance and weighed. The cap was removed and placed next to the bottle on the balance. The mass of the cap, bottle and its contents was monitored. The results are shown in the graph. The experiment was conducted at 25°C and 101.3 kPa. Assume that no evaporation has occurred.



- (a) Identify the gas released.

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- (b) Calculate the volume of the gas released.

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Question 24 (4 marks)

Assess the impact of atomic absorption spectroscopy (AAS) on the scientific understanding of the effects of trace elements.

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Centre Number

Section I – Part B (continued)

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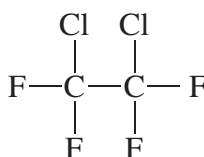
Student Number

Marks

Question 25 (6 marks)

- (a) What is the systematic name of the CFC in the diagram?

1



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- (b) Identify the bonding within ozone, using a Lewis electron-dot diagram.

2

- (c) Discuss how CFCs damage the ozone layer, using relevant equations.

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Question 26 (5 marks)

Water can be described as either ‘hard’ or ‘soft’.

- (a) Describe a test you have used to determine whether a given sample of water is ‘hard’ or ‘soft’. **2**

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- (b) A sample of hard water contains $6 \times 10^{-4} \text{ mol L}^{-1}$ of magnesium carbonate. **3**

Calculate the mass, in mg, of magnesium carbonate in 150 mL of this sample.

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Question 27 (5 marks)

Describe the physical and chemical processes needed to purify and sanitise a town water supply.

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Chemistry

Section II

25 marks

Attempt ONE question from Questions 28–32

Allow about 45 minutes for this section

Answer the question in a writing booklet. Extra writing booklets are available.

Show all relevant working in questions involving calculations.

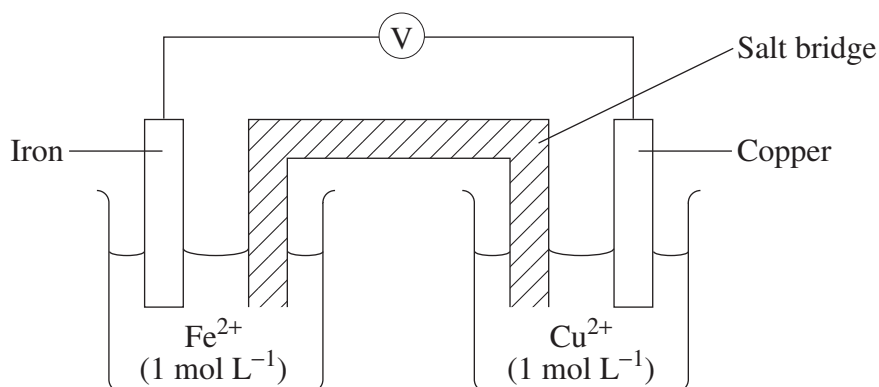
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Question 31 The Chemistry of Art	29
Question 32 Forensic Chemistry	30

Question 28 — Industrial Chemistry (25 marks)

- (a) (i) Define *saponification*. **1**
- (ii) Account for the cleaning action of soap. **3**
- (b) One of the reactions used to form sulfuric acid is the reaction of oxygen with sulfur dioxide under equilibrium conditions to form sulfur trioxide. **4**
- Before the reaction, the concentration of sulfur dioxide was 0.06 mol L^{-1} and the concentration of oxygen was 0.05 mol L^{-1} . After equilibrium was reached, the concentration of sulfur trioxide was 0.04 mol L^{-1} .
- Calculate the equilibrium constant, K , for the reaction. Show relevant working.
- (c) (i) Use a chemical equation to describe what happens when sulfuric acid is added to water in a laboratory. **2**
- (ii) Describe the use of sulfuric acid as an oxidising agent, as a dehydrating agent and as a means of precipitating sulfates. Use chemical equations to illustrate your answer. **3**
- (d) During your practical work, you performed a first-hand investigation involving an equilibrium reaction.
- (i) Outline the procedure you used. **2**
- (ii) Explain how you analysed the equilibrium reaction qualitatively. **4**
- (e) Evaluate changes in industrial production methods for sodium hydroxide. **6**

Question 29 — Shipwrecks and Salvage (25 marks)

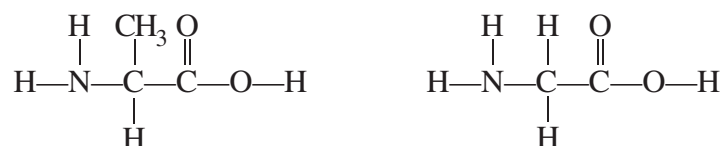
- (a) (i) Name the type of electrochemical cell that produces a spontaneous reaction. **1**
- (ii) Calculate the voltage required to operate the cell shown in the diagram as an electrolytic cell, showing relevant half-equations in your working. **3**



- (b) Describe how the work of early scientists increased our understanding of electron transfer reactions. **4**
- (c) (i) Name ONE method for removing salt from an artefact recovered from a wreck. **1**
- (ii) Explain, using an example, chemical procedures used to clean and preserve artefacts from wrecks. **4**
- (d) During your practical work you performed a first-hand investigation to compare and describe the rate of corrosion of materials in different acidic and neutral solutions.
- (i) Outline the procedure used. **2**
- (ii) It is hypothesised that acidic environments accelerate the corrosion of shipwrecks. **4**
- Explain how data obtained from the procedure in part (d) (i) does or does not support this hypothesis.
- (e) Analyse the effect of ocean depth on corrosion of metallic objects. **6**

Question 30 — The Biochemistry of Movement (25 marks)

- (a) (i) To what class of compounds do these molecules belong? **1**



- (ii) Using the two compounds in part (a) (i), describe the formation of a peptide bond, illustrating your answer with a chemical equation. Clearly indicate the peptide bond. **3**
- (b) Explain why triacylglycerols (TAGs) are an important energy store for humans. **4**
- (c) (i) Name the acid that builds up in muscles in anaerobic respiration. **1**
- (ii) Discuss the characteristics of muscle cells used for light exercise and those of muscle cells used for heavy exercise. **4**
- (d) During your practical work you performed a first-hand investigation to demonstrate the effect of various factors on the reaction of a named enzyme.
- (i) Name the enzyme and outline the procedure used to demonstrate the effect of ONE of these factors on the reaction. **2**
- (ii) Explain how the results obtained from the experiment in part (d) (i) can be used to indicate changes in the structure of the enzyme investigated. **4**
- (e) Discuss the oxidation of fatty acids as an inhibitor of the conversion of pyruvate to acetyl CoA, with reference to the dietary needs of sprinting athletes. **6**

Question 31 — The Chemistry of Art (25 marks)

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|-----|-------|---|----------|
| (a) | (i) | Identify the metal ion that will produce a yellow colour in a flame test. | 1 |
| | (ii) | Explain how some metal ions produce a characteristic colour in a flame. | 3 |
| (b) | | Using an example from medicine or biological research, describe the bonding in a coordination complex. | 4 |
| (c) | (i) | Explain what is meant by a reflectance spectrum. | 2 |
| | (ii) | Outline how infrared light and ultraviolet light are used to determine the chemical composition of pigments. | 3 |
| (d) | | During your practical work you performed a first-hand investigation to determine the oxidising strength of potassium permanganate. | |
| | (i) | State the electronic configuration of manganese in terms of subshells. | 1 |
| | (ii) | Outline the procedure used to determine the oxidising strength of potassium permanganate. | 2 |
| | (iii) | Using your results from the procedure in part (d) (ii), justify a conclusion about the oxidising strength of potassium permanganate. Use half-equations in your answer. | 3 |
| (e) | | Analyse trends in the physical properties of the first transition series. | 6 |

	Marks
Question 32 — Forensic Chemistry (25 marks)	
(a) (i) Identify the functional group in glycerol.	1
(ii) Compare the reactions of both glycerol and 1-propanol when they react with cold dilute KMnO_4 .	3
(b) Discuss the value of electron spectroscopy and scanning tunnelling microscopy in the analysis of small samples in forensic chemistry.	4
(c) (i) What class of compounds is used to break proteins into fragments of different lengths?	1
(ii) Describe the processes of electrophoresis and chromatography in separating organic compounds.	4
(d) During your practical work you performed a first-hand investigation to describe the emission spectrum of sodium.	
(i) Name the piece of equipment you used to analyse the emission spectrum of sodium in the laboratory.	1
(ii) Outline the procedure that you used in this investigation.	2
(iii) Explain how the emission spectrum was produced.	3
(e) Discuss the uses of DNA analysis in forensic chemistry.	6

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DATA SHEET

Avogadro's constant, N_A	$6.022 \times 10^{23} \text{ mol}^{-1}$
Volume of 1 mole ideal gas: at 101.3 kPa (1.00 atm) and	
at 273 K (0°C)	22.41 L
at 298 K (25°C)	24.47 L
Ionisation constant for water at 298 K (25°C), K_w	1.0×10^{-14}
Specific heat capacity of water	$4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$

Some useful formulae

$$\text{pH} = -\log_{10} [\text{H}^+]$$

$$\Delta H = -m C \Delta T$$

Some standard potentials

$\text{K}^+ + \text{e}^-$	$\rightleftharpoons$	K(s)	-2.94 V
$\text{Ba}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ba(s)	-2.91 V
$\text{Ca}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ca(s)	-2.87 V
$\text{Na}^+ + \text{e}^-$	$\rightleftharpoons$	Na(s)	-2.71 V
$\text{Mg}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Mg(s)	-2.36 V
$\text{Al}^{3+} + 3\text{e}^-$	$\rightleftharpoons$	Al(s)	-1.68 V
$\text{Mn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Mn(s)	-1.18 V
$\text{H}_2\text{O} + \text{e}^-$	$\rightleftharpoons$	$\frac{1}{2}\text{H}_2(\text{g}) + \text{OH}^-$	-0.83 V
$\text{Zn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Zn(s)	-0.76 V
$\text{Fe}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Fe(s)	-0.44 V
$\text{Ni}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ni(s)	-0.24 V
$\text{Sn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Sn(s)	-0.14 V
$\text{Pb}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Pb(s)	-0.13 V
$\text{H}^+ + \text{e}^-$	$\rightleftharpoons$	$\frac{1}{2}\text{H}_2(\text{g})$	0.00 V
$\text{SO}_4^{2-} + 4\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	$\text{SO}_2(\text{aq}) + 2\text{H}_2\text{O}$	0.16 V
$\text{Cu}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Cu(s)	0.34 V
$\frac{1}{2}\text{O}_2(\text{g}) + \text{H}_2\text{O} + 2\text{e}^-$	$\rightleftharpoons$	2OH^-	0.40 V
$\text{Cu}^+ + \text{e}^-$	$\rightleftharpoons$	Cu(s)	0.52 V
$\frac{1}{2}\text{I}_2(\text{s}) + \text{e}^-$	$\rightleftharpoons$	I^-	0.54 V
$\frac{1}{2}\text{I}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	I^-	0.62 V
$\text{Fe}^{3+} + \text{e}^-$	$\rightleftharpoons$	Fe^{2+}	0.77 V
$\text{Ag}^+ + \text{e}^-$	$\rightleftharpoons$	Ag(s)	0.80 V
$\frac{1}{2}\text{Br}_2(\text{l}) + \text{e}^-$	$\rightleftharpoons$	Br^-	1.08 V
$\frac{1}{2}\text{Br}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	Br^-	1.10 V
$\frac{1}{2}\text{O}_2(\text{g}) + 2\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	H_2O	1.23 V
$\frac{1}{2}\text{Cl}_2(\text{g}) + \text{e}^-$	$\rightleftharpoons$	Cl^-	1.36 V
$\frac{1}{2}\text{Cr}_2\text{O}_7^{2-} + 7\text{H}^+ + 3\text{e}^-$	$\rightleftharpoons$	$\text{Cr}^{3+} + \frac{7}{2}\text{H}_2\text{O}$	1.36 V
$\frac{1}{2}\text{Cl}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	Cl^-	1.40 V
$\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^-$	$\rightleftharpoons$	$\text{Mn}^{2+} + 4\text{H}_2\text{O}$	1.51 V
$\frac{1}{2}\text{F}_2(\text{g}) + \text{e}^-$	$\rightleftharpoons$	F^-	2.89 V

PERIODIC TABLE OF THE ELEMENTS

PERIODIC TABLE OF THE ELEMENTS										KEY									
1 H 1.008 Hydrogen		4 Be 9.012 Beryllium		Atomic Number		79 Au 197.0 Gold		Symbol of element		8 O 16.00 Oxygen		9 F 19.00 Fluorine		2 He 4.003 Helium					
3 Li 6.941 Lithium		12 Mg 24.31 Magnesium		Atomic Weight		197.0 Gold		Name of element		6 C 12.01 Carbon		7 N 14.01 Nitrogen		10 Ne 20.18 Neon					
11 Na 22.99 Sodium		18 Ar 39.95 Argon								14 Si 28.09 Silicon		15 P 30.97 Phosphorus		17 Cl 35.45 Chlorine					
19 K 39.10 Potassium		36 Kr 83.80 Krypton								31 Ga 69.72 Gallium		33 As 74.92 Arsenic		35 Br 79.90 Bromine					
37 Rb 85.47 Rubidium		54 Xe 131.3 Xenon								49 In 114.8 Indium		51 Sb 121.8 Antimony		53 I 126.9 Iodine					
55 Cs 132.9 Caesium		86 Rn [222.0] Radon								81 Tl 204.4 Thallium		83 Bi 209.0 Bismuth		85 At [210.0] Astatine					
87 Fr [223.0] Francium		118 Uuo — Ununoctium								113 Uuq — Ununquadium		115 Uup — Ununpentium		117 Uus — Ununseptium					

Lanthanides

57 La 138.9 Lanthanum	58 Ce 140.1 Cerium	59 Pr 140.9 Praseodymium	60 Nd 144.2 Neodymium	61 Pm [146.9] Promethium	62 Sm 150.4 Samarium	63 Eu 152.0 Europium	64 Gd 157.3 Gadolinium	65 Tb 158.9 Terbium	66 Dy 162.5 Dysprosium	67 Ho 164.9 Holmium	68 Er 167.3 Erbium	69 Tm 168.9 Thulium	70 Yb 173.0 Ytterbium	71 Lu 175.0 Lutetium
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Actinides

89 Ac [227.0] Actinium	90 Th 232.0 Thorium	91 Pa 231.0 Protactinium	92 U 238.0 Uranium	93 Np [237.0] Neptunium	94 Pu [239.1] Plutonium	95 Am [241.1] Americium	96 Cm [244.1] Curium	97 Bk [249.1] Berkelium	98 Cf [252.1] Californium	99 Es [252.1] Einsteinium	100 Fm [257.1] Fermium	101 Md [258.1] Mendelevium	102 No [259.1] Nobelium	103 Lr [262.1] Lawrencium
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Where the atomic weight is not known, the relative atomic mass of the most common radioactive isotope is shown in brackets.
The atomic weights of Np and Tc are given for the isotopes ²³⁷Np and ⁹⁹Tc.

**2002 HSC Notes from
the Marking Centre
Chemistry**

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2002 HSC NOTES FROM THE MARKING CENTRE

CHEMISTRY

Introduction

This document has been produced for the teachers and candidates of the Stage 6 course in Chemistry. It provides comments with regard to responses to the 2002 Higher School Certificate Examination, indicating the quality of candidate responses and highlighting the relative strengths and weaknesses of the candidature in each section and each question.

It is essential for this document to be read in conjunction with the relevant syllabus, the 2002 Higher School Certificate Examination, the Marking Guidelines and other support documents which have been developed by the Board of Studies to assist in the teaching and learning of Chemistry.

General Comments

In 2002, 8867 candidates attempted the chemistry examination.

This year candidates seemed to have a greater awareness of the importance of the key verbs in the examination. There was a slight improvement in the ability of candidates to successfully respond to the higher level verbs in extended response questions.

There was a significant improvement in the setting out of calculations so that responses could be more easily understood.

Candidates need to be more aware of the full requirements of questions.

The variation in the quality of responses seems to suggest that candidates who have actively planned and performed a first-hand investigation are more likely to have retained knowledge and understanding of the experience than those who have rote learnt an expected answer.

Teachers and candidates should be aware that examiners may write questions that address the syllabus outcomes in a manner that requires candidates to respond by integrating their knowledge, understanding and skills developed through studying the course. This reflects the fact that the knowledge, understanding and skills developed through the study of discrete sections should accumulate to a more comprehensive understanding than may be described in each section separately.

Section I – Core**Part A – Multiple choice**

Question	Correct Response
1	A
2	B
3	C
4	A
5	D
6	C
7	B
8	A

Question	Correct Response
9	D
10	D
11	B
12	D
13	C
14	C
15	A

Part B**General Comments**

Overall, the candidates' responses indicated that the majority had a grasp of chemical concepts, appropriate for HSC candidates. Candidates need to be aware that the answer space allocated is a guide to the length of the required response. A succinct answer, within the allocated number of lines, is all that is needed to gain full marks for a question.

Specific Comments**Question 16**

Most candidates' answers indicated that they had carried out a first-hand investigation.

However, a significant number of candidates responded with reagents such as ethene, ethane and chlorine which indicated that the candidate had not carried out a first-hand investigation. The omission of equipment used or amounts of chemicals used also indicated lack of first-hand experience. The procedure was not done well by candidates who only had knowledge of the expected results.

Question 17

Explanations of the similar physical properties of alkanes and corresponding alkenes were poorly answered while the difference in chemical properties was mostly answered correctly.

Question 18

- (a) More than 90% of candidates correctly named the type of polymerisation.

- (b) Many candidates appear to have memorised an answer about biopolymers without necessarily understanding the chemistry involved. Where a question asks for plural responses, as indicated by developments and biopolymers, candidates must supply more than one example in their answer. A question asking about current developments calls for a response describing recent developments.

Question 19

Most candidates indicated a reasonable understanding of nuclear chemistry. A significant number of candidates wrote excessively long answers. Such long answers are likely to have missed the point of the question or provide responses that contain contradictions.

- (a) In this part most candidates wrote too much, going well beyond the allocated space. Many candidates erroneously included electrons in their answer to a question on nuclear stability.
- (b) The majority of candidates interpreted the graph well.

Question 20

- (a) Candidates who responded by naming a chemical that was a base usually obtained a mark. Naming a commercial product mixture was not a satisfactory response.
- (b) The assessment of the validity of a conclusion was not well done. Some candidates confused reliability and validity. Others discussed the validity of the experimental design rather than validity of the conclusion. The eight lines of space provided were adequate to achieve full marks.

Some candidates did not appreciate the concept of pH range, wanting to average the range figures.

Question 21

It was encouraging to see underlining of key words and other evidence of planning by candidates before responding to this question. Most candidates demonstrated knowledge of the impacts of industrial sources and of sulphur dioxide and nitrogen oxides on the environment. Equations given were generally appropriate and accurate. Few candidates demonstrated in their answer the ability to make a value judgement as is required when asked to 'evaluate'.

Question 22

- (a) Most candidates correctly calculated the pH of the hydrochloric solution.
- (b) Many candidates used general terms like more or less acidic in their answer rather than comparing the pH values.
- (c) Generally well answered but some candidates confused higher pH with higher hydrogen ion concentration.

Question 23

- (a) The majority of candidates answered this question correctly.
- (b) The calculation was generally well done, many candidates achieving full marks.

Question 24

Candidates generally had difficulty interpreting this question. It was encouraging to see the key words underlined as part of the candidate's preparation for response. However, many responses did not show evidence of specific knowledge of trace elements and their effects. A good understanding of AAS was demonstrated, but many failed to assess its impact in terms of improvement over other techniques.

Question 25

- (a) Many candidates had difficulty naming the compounds.
- (b) Few candidates could correctly draw the required diagram and identify the two types of bonding present.
- (c) Candidates who wrote about chlorine ions, chloride ions or chlorine molecules could not achieve the three marks. It is the chlorine atom or chlorine free radical released by ultraviolet radiation from a CFC molecule that reacts with ozone.

Question 26

- (a) This was poorly answered in terms of the skills associated with outcomes H13 and H14, describing tests and detailing anticipated results. Many candidates named a test instead of describing it.
- (b) The calculation was quite well done.

Question 27

Candidates responded well to this question with hardly any non-attempts and with most candidates able to describe physical and chemical processes.

Section II – Options

Specific Comments

Question 28 – Industrial Chemistry

- (a) (i) Very few candidates provided the required detail, most simply writing ‘soap making’ – insufficient to gain a mark in an option where they had studied the process in detail.
- (ii) This question required the candidates to describe the process using details of the structure of soap particles and the attraction of different parts of that structure for polar water and non-polar substances.
- (b) Many candidates did not realise that they had to process the information provided to determine the concentrations at equilibrium.
- (c) (i) A significant number of candidates were unable to write a balanced equation.
- (ii) Candidates made better use of equations in this part but many descriptions included incorrect or contradictory statements.
- (d) (i) A surprising number of candidates claimed to have carried out the Haber process as a first-hand investigation, evidence of a rote learned procedure, rather than having carried out a first-hand investigation.
- (ii) In this part candidates were expected to be able to apply core conceptual understanding of Le Chatelier’s principle to analyse an equilibrium that had been performed as a first-hand investigation in the option. Responses were spread fairly uniformly over the full range of available marks.
- (e) This question illustrates the point made in the last paragraph in the General Comments on page 5. The syllabus requires candidates to analyse the technical and environmental difficulties associated with the three main processes used to extract sodium hydroxide. The candidates needed to make a value judgement to answer this question completely.

Question 29 – Shipwrecks and Salvage

- (a) (i) A sizeable minority of candidates gave an example of an electrochemical cell rather than naming the type as required.
- (ii) Most candidates identified the appropriate half-equations and performed the correct calculation without recognising that a slightly higher voltage than that calculated was needed.
- (b) Some candidates responded with the historical context of proton-transfer reactions which is not part of this option. Most correctly responded with descriptions related to electron transfer reactions, but there was a lack of understanding of the work of Galvani and Volta.

- (c) (i) Most candidates could name a suitable chemical or physical method.
- (ii) Candidates often showed confusion between the methods required for cleaning, stabilising and preserving. Chemical procedures were also often confused with physical/mechanical cleaning of artefacts. Candidates should recognise the need to provide a response that shows a causal link to address the ‘how and/or why’ nature of an ‘explain’ question.
- (d) (i) Generally well done but most candidates did not suggest a method to measure the amount of corrosion. Some candidates thought that salt water was acidic.
- (ii) Candidates were required to describe the results of their first-hand investigation and relate how these supported or did not support the hypothesis. Most candidates did not distinguish between the observed results and the conclusion for their practical work. There were indications that some candidates had not actually done the required practical work.
- (e) Generally well done, however, some responses did not show the links between facts as is required to meet the demands of a question asking candidates to ‘analyse’.

Question 30 – Biochemistry of Movement

- (a) (i) Most candidates answered the question correctly.
- (ii) Answered well.
- (b) Many candidates were aware that TAGs provide more energy per gram than carbohydrates but few explained that TAGs could be converted to carbohydrates and vice versa.
- (c) (i) Most candidates answered this correctly.
- (ii) The responses indicated a high level of understanding.
- (d) (i) The responses indicated that most candidates had performed a first-hand investigation. Some, however, confused the enzyme renin with the product junket. Few candidates correctly used a control or described a method of collecting reliable results.
- (ii) Many candidates did not give the results of their practical procedure. Those that did, often had difficulty linking their practical observations with changes in the structure of the enzyme. Most candidates demonstrated some knowledge of secondary and tertiary structure and were able to discuss denaturation as a change in structure/shape that could destroy enzyme activity.
- (e) Many candidates stated that fatty acid oxidation produces acetyl CoA but no candidate made the connection with the dietary needs of sprinters.

Question 31 – Chemistry of Art

- (a) (i) Generally well answered.
- (ii) Generally well answered.
- (b) Candidates scored well by focusing on the nature of the bonding rather than just the example to be used in describing the bonding.
- (c) (i) Many candidates repeated the question with a slight embellishment such as ‘when light is reflected a reflectance spectrum is produced’. This was insufficient to gain any mark.
- (ii) Most candidates did not know how infrared and ultraviolet light are used, frequently mentioning visible light or explaining reflectance spectra again. The idea of relating the data obtained to standards was understood by only a small number of candidates.
- (d) (i) The responses indicated a high level of understanding of this content.
- (ii) It was obvious to the markers that many candidates had not performed the practical work. Frequently candidates would choose combinations of reagents that would not react yet claimed that they did. The idea of ‘oxidising strength’ was frequently confused with ‘concentration’ of potassium permanganate.
- (iii) Very few candidates identified any results, instead common errors included justifying a conclusion and then exploring the use of half equations, half equations involving hydrogen ions used where there had been no mention of acid conditions in part (ii), reversible arrows being left in half equations, and transcription errors.
- (e) Most candidates gained some marks. Candidates showed some lack of understanding of how the physical properties identified related to the electron arrangement in the 3d subshell. Atomic radius and melting point trends were handled best while density and magnetism trends were less well understood.

Question 32 – Forensic Chemistry

- (a) (i) The majority of candidates answered this correctly. Common errors were hydroxide ion, hydroxide or alcohol.
- (ii) Many candidates answered this part correctly.
- (b) Many responses were vague and could have been about any scientific instrument. Few candidates gave applications or examples of the use of the techniques.
- (c) (i) Few candidates answered this correctly. Many capable candidates lost the mark by being too specific and negating the enzyme part of their response by specifying restriction enzymes.
- (ii) A large number of candidates showed a particularly good understanding of electrophoresis. Responses on chromatography were not as detailed or accurate.

- (d) (i) Many candidates did not respond with ‘spectroscope’ even though they described its use correctly in (d)(ii).
- (ii) Generally well done – candidates had obviously attempted this practical work. A significant number of candidates, however, described a flame test or failed to accurately describe what they were looking for in a spectrum.
- (iii) Answered well by the majority of candidates.
- (e) Candidates should be advised to plan their responses before writing their final answer. Many responses were disjointed or repetitive lists of examples. Selected examples were often dealt with superficially and failed to show an understanding of either the chemistry of DNA or the processes involved in its use in forensic chemistry.

Chemistry

2002 HSC Examination Mapping Grid

Question	Marks	Content	Syllabus outcomes
Section I Part A			
1	1	9.2.3	H4, H8
2	1	9.2.3	H8
3	1	9.2.4	H1, H3
4	1	9.2.4	H7, H13
5	1	9.2.4	H8, H14
6	1	9.3.4	H10
7	1	9.3.4	H1, H2
8	1	9.3.4	H11, H12
9	1	9.3.5	H8, H10, H11, H12
10	1	9.3.5	H9, H10
11	1	9.4.1	H6, H8, H9, H10
12	1	9.4.1	H13, H14
13	1	9.4.2	H1, H4
14	1	9.4.2	H2, H7, H8, H14
15	1	9.4.3	H2, H3, H8, H10, H14
16 (a)	1	9.2.1	H9
16 (b)	2	9.2.1	H9, H11, H13
16 (c)	3	9.2.1	H9, H11, H13, H14
17	3	9.2.1	H1, H3, H9, H13
18 (a)	1	9.2.2	H9
18 (b)	5	9.2.2	H3, H4, H9
19 (a)	2	9.2.5	H1, H2, H6
19 (b)	3	9.2.5	H1, H2, H6
20 (a)	1	9.3.1	H6, H8
20 (b)	3	9.3.1	H1, H2, H3, H10, H11, H12, H13, H14
21	7	9.3.2	H1, H3, H4, H5, H6, H7, H8, H10, H13
22 (a)	1	9.3.3	H10, H13
22 (b)	2	9.3.3	H2, H6, H10, H13
22 (c)	2	9.3.3	H2, H6, H10, H13, H14
23 (a)	1	9.3.2	H6, H8, H10
23 (b)	3	9.3.2	H6, H7, H10, H13, H14

Question	Marks	Content	Syllabus outcomes
24	4	9.4.3	H1, H3, H7
25 (a)	1	9.4.4	H9, H13
25 (b)	2	9.4.4	H2, H6, H13
25 (c)	3	9.4.4	H2, H4, H6, H7, H8, H9, H10, H13
26 (a)	2	9.4.5	H3, H4, H8, H11, H13
26 (b)	3	9.4.5	H10, H13
27	5	9.4.5	H3, H4, H5, H8, H13
Section II — Industrial Chemistry			
28 (a) (i)	1	9.5.5	H2, H3, H4, H9, H10
28 (a) (ii)	3	9.5.5	H2, H3, H4, H9, H10
28 (b)	4	9.5.2	H2, H3, H10, H12, H13
28 (c) (i)	2	9.5.3	H6, H7, H8, H10, H13
28 (c) (ii)	3	9.5.3	H1, H2, H3, H8, H9, H10, H13
28 (d) (i)	2	9.5.2	H2, H3, H10, H13
28 (d) (ii)	4	9.5.2	H2, H3, H10, H13
28 (e)	6	9.5.4	H1, H2, H3, H4, H7, H8
Section II — Shipwrecks and Salvage			
29 (a) (i)	1	9.6.3	H2, H3, H7, H8
29 (a) (ii)	3	9.6.3	H2, H3, H7, H8, H10, H13
29 (b)	4	9.6.1	H1, H2, H3, H4, H7
29 (c) (i)	1	9.6.7	H1, H3, H4, H5, H6, H8
29 (c) (ii)	4	9.6.7	H1, H3, H4, H5, H6, H8, H11, H13
29 (d) (i)	2	9.6.6	H3, H6, H8, H10, H11, H13, H14
29 (d) (ii)	4	9.6.6	H3, H6, H8, H10, H11, H13, H14
29 (e)	6	9.6.5, 9.6.6	H2, H3, H4, H5, H6, H7, H8, H10, H13
Section II — The Biochemistry of Movement			
30 (a) (i)	1	9.7.4	H9, H13
30 (a) (ii)	3	9.7.4	H6, H9, H10, H13
30 (b)	4	9.7.3	H2, H3, H4, H7, H13
30 (c) (i)	1	9.7.11	H7, H9, H10, H13
30 (c) (ii)	4	9.7.6	H2, H4, H8, H13
30 (d) (i)	2	9.7.4	H2, H3, H8, H9, H11, H13

Question	Marks	Content	Syllabus outcomes
30 (d) (ii)	4	9.7.4	H2, H3, H7, H8, H9, H11, H13, H14
30 (e)	6	9.7.7, 9.7.8, 9.7.11	H2, H3, H4, H8, H9, H13
Section II — The Chemistry of Art			
31 (a) (i)	1	9.8.3	H2, H3, H4, H7, H11, H13
31 (a) (ii)	3	9.8.3, 9.8.4	H2, H3, H4, H7, H11, H13
31 (b)	4	9.8.7	H2, H3, H4, H6, H10, H13
31 (c) (i)	2	9.8.3	H2, H3, H7, H13
31 (c) (ii)	3	9.8.3	H2, H3, H4, H7, H13
31 (d) (i)	1	9.8.4, 9.8.5	H2, H3, H6, H7, H13
31 (d) (ii)	2	9.8.5	H2, H6, H11
31 (d) (iii)	3	9.8.5	H2, H6, H7, H8, H9, H10, H13, H14
31 (e)	6	9.8.5, 9.8.6	H2, H6, H13, H14
Section II — Forensic Chemistry			
32 (a) (i)	1	9.9.3	H2, H9
32 (a) (ii)	3	9.9.3	H8, H9, H13
32 (b)	4	9.9.6	H1, H3, H4, H12, H13
32 (c) (i)	1	9.9.4	H3, H4, H8, H9, H13
32 (c) (ii)	4	9.9.4	H1, H3, H4, H13
32 (d) (i)	1	9.9.7	H1, H2, H3, H7
32 (d) (ii)	2	9.9.7	H1, H2, H3, H7, H11, H13
32 (d) (iii)	3	9.9.7	H1, H2, H3, H4, H7, H11, H13
32 (e)	6	9.9.5	H1, H3, H4, H5, H13

2002 HSC Chemistry Marking Guidelines

Section I Part B

Question 16 (a)

Outcomes assessed: H9

MARKING GUIDELINES

Criteria	Marks
• States the name of a plausible alkene	1

Question 16 (b)

Outcomes assessed: H9, H11, H13

MARKING GUIDELINES

Criteria	Marks
• Gives a correct procedure that could be used to compare the reactivity of the chosen alkene with its corresponding alkane, including the reagent	2
• Gives a partially correct procedure that could be used to compare the reactivity of the chosen alkene with its corresponding alkane	1
OR	
• Correctly names the reagent for the procedure	

Question 16 (c)

Outcomes assessed: H9, H11, H13, H14

MARKING GUIDELINES

Criteria	Marks
• Results described correctly, with correct equation	3
• Results described correctly for one reaction, with equation OR • Results described correctly	2
• Results described correctly for one reaction OR • Correct equation	1

Question 17

Outcomes assessed: H1, H3, H9, H13

MARKING GUIDELINES

Criteria	Marks
• Explains the similarities of physical properties and the difference in chemical properties of alkanes and alkenes	3
• Explains a physical property of alkanes and alkenes OR • Explains a chemical property of alkanes and alkenes OR • Describes both a physical and a chemical property of alkanes and alkenes	2
• Identifies the polarity of the molecules OR • Identifies the single bonds in alkanes and double bonds in alkenes OR • Identifies a similar physical property of alkanes and alkenes OR • Identifies a difference in chemical reactivity between alkanes and alkenes	1

Question 18 (a)

Outcomes assessed: H9

MARKING GUIDELINES

Criteria	Marks
• Correct answer given	1

Question 18 (b)

Outcomes assessed: H3, H4, H9

MARKING GUIDELINES

Criteria	Marks
- Describes accurately current developments in the use of biopolymers AND - Makes a valid judgement of the value of those developments	4–5
- Describes accurately current developments in the use of biopolymers OR - Accounts for the current developments in the use of biopolymers OR - Describes accurately ONE current development in the use of biopolymers and accounts for a benefit/limitation in the use of biopolymers	2–3
- Names a biopolymer OR - Gives a use for a biopolymer OR - Identifies a benefit/limitation for biopolymers	1

Question 19 (a)

Outcomes assessed: H1, H2, H6

MARKING GUIDELINES

Criteria	Marks
Correctly identifies the features that make a nucleus unstable	2
- Identifies that ‘protons’ and ‘neutrons’ are found in nucleus OR - Correctly describes one condition under which a nucleus is unstable	1

Question 19 (b)

Outcomes assessed: H1, H2, H6

MARKING GUIDELINES

Criteria	Marks
Describes alpha and beta decay using examples from the flow diagram	3
- Describes alpha decay using an example from the flow diagram OR - Describes beta decay using an example from the flow diagram OR - Describes alpha and beta decay	2
Describes either alpha or beta decay	1

Question 20 (a)

Outcomes assessed: H6, H8

MARKING GUIDELINES

Criteria	Marks
Any correct household base	1

Question 20 (b)

Outcomes assessed: H1, H2, H3, H10, H11, H12, H13, H14

MARKING GUIDELINES

Criteria	Marks
Correctly concludes solutions <i>A</i> and <i>B</i> must be basic, and concludes, based on correct reasoning, that solution <i>C</i> may be in the range basic to acidic and assesses that the conclusion may be or may not be correct	3
Correctly concludes solution <i>A</i> and <i>B</i> must be basic and concludes, based on correct reasoning, that solution <i>C</i> may be in the range basic to acidic	2
- Correctly concludes solutions <i>A</i> or <i>B</i> must be basic OR - Makes a correct assessment without any explanation	1

Question 21

Outcomes assessed: H1, H3, H4, H5, H6, H7, H8, H10, H13

MARKING GUIDELINES

Criteria	Marks
• Demonstrates an extensive knowledge of the industrial production of nitrogen oxides and sulfur dioxide • Demonstrates an extensive knowledge of impacts on the environment • Includes examples and chemical equations • Provides an evaluation of these impacts	7
• Demonstrates an extensive knowledge of the industrial production of nitrogen oxides and sulfur dioxide • Demonstrates an extensive knowledge of impacts on the environment • Includes chemical equation(s) OR • Demonstrates a sound knowledge of impacts • Uses chemical equations and gives a limited evaluation • Demonstrates a sound knowledge of the industrial production of nitrogen oxides and sulfur dioxide	5–6
• Demonstrates a sound knowledge of industrial production of sulfur dioxide and nitrogen oxides • Demonstrates limited knowledge of environmental impacts OR • Demonstrates a sound knowledge of environmental impacts • Demonstrates limited knowledge of industrial production of sulfur dioxide and nitrogen oxides	3–4
• Names oxides of nitrogen or sulfur dioxide or gives formulas AND/OR • Names industrial sources of a nitrogen oxide or sulfur dioxide AND/OR • States environmental effects of an oxide of nitrogen or of sulfur dioxide	1–2

Question 22 (a)

Outcomes assessed: H10, H13

MARKING GUIDELINES

Criteria	Marks
Correctly calculates pH	1

Question 22 (b)

Outcomes assessed: H2, H6, H10, H13

MARKING GUIDELINES

Criteria	Marks
- Identifies that sulfuric acid has a lower pH than hydrochloric acid - Justifies answer in terms of the diprotic nature of sulfuric acid	2
- Identifies that sulfuric acid has a lower pH than hydrochloric acid OR - States that sulfuric acid is diprotic whereas hydrochloric acid is monoprotic	1

Question 22 (c)

Outcomes assessed: H2, H6, H10, H13, H14

MARKING GUIDELINES

Criteria	Marks
- Distinguishes the acids in terms of their 'strength' and hence their $[H^+]$ - Correctly relates pH with $[H^+]$	2
- Distinguishes the acids in terms of their 'strength' and hence their $[H^+]$ OR - Correctly relates pH with $[H^+]$	1

Question 23 (a)

Outcomes assessed: H6, H8, H10

MARKING GUIDELINES

Criteria	Marks
Names the gas correctly	1

Question 23 (b)

Outcomes assessed: H6, H7, H10, H13, H14

MARKING GUIDELINES

Criteria	Marks
• Correctly calculates the volume of gas, including units and showing relevant working	3
• Correct method used, but has a mistake*	2
• Correct number of moles of gas calculated	1

*Mistakes could include: transcription errors, using wrong formula mass, using 22.4L, missing units.

Question 24

Outcomes assessed: H1, H3, H7

MARKING GUIDELINES

Criteria	Marks
• Describes how the use of AAS has changed analytical procedures and assesses the impact of AAS on scientific understanding of the effects of trace elements	4
• Describes how the use of AAS has changed analytical procedures AND/OR • Describes how the use of AAS has enhanced scientific understanding of trace elements AND/OR • A good explanation of how AAS works AND/OR • Describes the effects of trace elements	2–3
• Identifies some trace elements measured using AAS OR • Describes AAS OR • Indicates an advantage of AAS	1

Question 25 (a)

Outcomes assessed: H9, H13

MARKING GUIDELINES

Criteria	Marks
• Correct answer given	1

Question 25 (b)

Outcomes assessed: H2, H6, H13

MARKING GUIDELINES

Criteria	Marks
Identifies the bonding using a Lewis electron-dot diagram	2
- Draws Lewis electron-dot diagram OR - Identifies the position of the coordinate covalent bond OR - Identifies the position of the double covalent bond	1

Question 25 (c)

Outcomes assessed: H2, H4, H6, H7, H8, H9, H10, H13

MARKING GUIDELINES

Criteria	Marks
Demonstrates extensive knowledge of the reactions of CFCs with ozone. Must use relevant equations	3
- Demonstrates extensive knowledge of the reactions without equations OR - Demonstrates a limited knowledge of reactions with equations	2
Demonstrates a limited knowledge of reactions or species present	1

Question 26 (a)

Outcomes assessed: H3, H4, H8, H11, H13

MARKING GUIDELINES

Criteria	Marks
Describes test with expected distinguishing results	2
- Describes test OR - Gives results without test	1

Question 26 (b)

Outcomes assessed: H10, H13

MARKING GUIDELINES

Criteria	Marks
• Gives correct answer and working	3
• Incorrect answer with some correct working	2
• Correct molar mass for MgCO_3 OR • Correct formula for conversion of moles/L to moles OR • Correct answer without working	1

Question 27

Outcomes assessed: H3, H4, H5, H8, H13

MARKING GUIDELINES

Criteria	Marks
• Demonstrates an extensive knowledge of both chemical and physical processes used to purify and sanitise a town water supply	5
• Demonstrates a thorough knowledge of both chemical and physical processes used to purify and sanitise a town water supply OR • Demonstrates an extensive knowledge of either chemical or physical processes used to purify and sanitise a town water supply	3–4
• Demonstrates a basic knowledge of both chemical and physical processes used to purify and sanitise a town water supply OR • Demonstrates a sound knowledge of either chemical or physical processes used to purify and sanitise a town water supply	1–2

Section II

Industrial Chemistry

Question 28 (a) (i)

Outcomes assessed: H2, H3, H4, H9, H10

MARKING GUIDELINES

Criteria	Marks
• Correctly defines the process	1

Question 28 (a) (ii)

Outcomes assessed: H2, H3, H4, H9, H10

MARKING GUIDELINES

Criteria	Marks
• States reasons for the cleaning action of soap	3
• Identifies water solubility of soaps AND • Identifies polar and non-polar ends of soaps	2
• Identifies water solubility of soaps OR • Identifies polar and non-polar ends of soaps	1

Question 28 (b)

Outcomes assessed: H2, H3, H10, H12, H13

MARKING GUIDELINES

Criteria	Marks
• Gives correct answer with working and equations	4
• Gives correct equilibrium values for gases AND • Gives correct equilibrium expression AND • Gives correct chemical equation	3
• Gives correct equilibrium expression AND • Gives correct chemical equation	2
• Gives correct equilibrium expression OR • Gives correct chemical equation	1

Question 28 (c) (i)

Outcomes assessed: H6, H7, H8, H10, H13

MARKING GUIDELINES

Criteria	Marks
Gives correct equation and states that ionisation is exothermic	2
- States that the reaction is exothermic OR - Gives correct equation	1

Question 28 (c) (ii)

Outcomes assessed: H1, H2, H3, H8,H9, H10, H13

MARKING GUIDELINES

Criteria	Marks
- States three correct reactions and gives at least two correct equations OR - Gives three correct equations	3
- States three correct reactions OR - States two correct reactions and gives one correct equation OR - Gives two correct equations	2
- States one correct reaction OR - Gives one correct equation	1

Question 28 (d) (i)

Outcomes assessed: H2, H3, H10, H13

MARKING GUIDELINES

Criteria	Marks
Demonstrates a detailed knowledge of the use of appropriate experimental procedures in an equilibrium reaction	1–2

Question 28 (d) (ii)

Outcomes assessed: H2, H3, H10, H13

MARKING GUIDELINES

Criteria	Marks
Demonstrates an extensive knowledge of equilibria by analysis of the effects of at least two factors	4
- Demonstrates a thorough knowledge of equilibria by analysis of the effects of at least two factors OR - Demonstrates an extensive knowledge by analysis of the effects of one factor	3
- Describes the effect of two factors on the equilibrium OR - Describes the effect of one factor on the equilibrium and mentions Le Chatelier's principle	2
- Describes the effect of one factor on the equilibrium OR - Mentions Le Chatelier's principle	1

Question 28 (e)

Outcomes assessed: H1, H2, H3, H4, H7, H8

MARKING GUIDELINES

Criteria	Marks
Demonstrates an extensive knowledge of the changes in the industrial production methods for NaOH and evaluates the changes	5–6
Demonstrates a thorough knowledge of the changes in the industrial production methods for NaOH	3–4
Demonstrates a limited knowledge of changes in the industrial production methods for NaOH	1–2

Section II

Shipwrecks and salvage

Question 29 (a) (i)

Outcomes assessed: H2, H3, H7, H8

MARKING GUIDELINES

Criteria	Marks
• Correctly identifies the name of the electrochemical cell	1

Question 29 (a) (ii)

Outcomes assessed: H2, H3, H7, H8, H10, H13

MARKING GUIDELINES

Criteria	Marks
• Indicates that the correct voltage for the electrolytic cell is greater than the galvanic cell, and calculates this voltage, and includes correct relevant equations	3
• Calculates the potential of the galvanic cell illustrated and writes correct equations	2
• Writes one correct equation OR • Calculates the potential of the galvanic cell	1

Question 29 (b)

Outcomes assessed: H1, H2, H3, H4, H7

MARKING GUIDELINES

Criteria	Marks
• Demonstrates a thorough knowledge of the work of early scientists which led to an understanding of electron transfer reactions	3–4
• Demonstrates a sound knowledge of the work of early scientists which led to an understanding of electron transfer reactions	2
• Demonstrates a basic knowledge of the work of early scientists which led to an understanding of electron transfer reactions	1

Question 29 (c) (i)

Outcomes assessed: H1, H3, H4, H5, H6, H8

MARKING GUIDELINES

Criteria	Marks
Correctly names a method for removing salt from an artefact	1

Question 29 (c) (ii)

Outcomes assessed: H1, H3, H4, H5, H6, H8, H11, H13

MARKING GUIDELINES

Criteria	Marks
Provides a thorough explanation, using an example, of chemical procedures to clean and preserve artefacts from wrecks	3–4
- Provides a thorough explanation of chemical procedures used to either clean or preserve an artefact from wrecks - Provides a sound knowledge of chemical procedures to clean and preserve an artefact from wrecks	2
- Identifies ONE chemical procedure that is used to either clean or preserve an artefact from a wreck - Correctly identifies an example of an artefact from a wreck	1

Question 29 (d) (i)

Outcomes assessed: H3, H6, H8, H10, H11, H13, H14

MARKING GUIDELINES

Criteria	Marks
Demonstrates a thorough knowledge of the use of appropriate experimental procedures for comparing and describing the rate of corrosion of materials in different acidic and neutral solutions	1–2

Question 29 (d) (ii)

Outcomes assessed: H3, H6, H8, H10, H11, H13, H14

MARKING GUIDELINES

Criteria	Marks
- Provides a thorough description of the results - Provides a thorough explanation of how the results do or do not support the hypothesis	3–4
- Provides a basic explanation of how the results do or do not support the hypothesis OR - Describes the results	1–2

Question 29 (e)*Outcomes assessed: H2, H3, H4, H5, H6, H7, H8, H10, H13***MARKING GUIDELINES**

Criteria	Marks
• Demonstrates an extensive knowledge of the factors that affect corrosion of metallic objects and how these factors are affected by ocean depth	5–6
• Demonstrates a sound knowledge of factors that affect corrosion of metallic objects and how these factors are affected by ocean depth	3–4
• Identifies how one factor, that can affect corrosion of a metallic object, varies with depth	2
• Identifies one factor that will affect the corrosion of metallic objects and that changes with depth	1

Section II

The Biochemistry of Movement

Question 30 (a) (i)

Outcomes assessed: H9, H13

MARKING GUIDELINES

Criteria	Marks
Correctly identifies class of compound	1

Question 30 (a) (ii)

Outcomes assessed: H6, H9, H10, H13

MARKING GUIDELINES

Criteria	Marks
- Demonstrates a thorough knowledge of peptide bond formation, using the correct equation - Describes elimination of water from carboxyl and amino groups and identifies the peptide band	3
- Demonstrates a sound knowledge of peptide bond formation describing the elimination of water from carboxyl and amino groups OR - Gives an equation for the peptide bond formation with an error not including the peptide bond	2
- Demonstrates a basic knowledge of peptide bond formation by identifying the peptide bond OR - Describing the role of amino acids in the formation of the bond OR - Providing a simple description of the peptide bond formation	1

Question 30 (b)

Outcomes assessed: H2, H3, H4, H7, H13

MARKING GUIDELINES

Criteria	Marks
- Demonstrates an extensive knowledge of TAG features and the reasons why these features make TAGs important energy store AND - Demonstrates that TAGs can be converted to and from other energy sources as needed	4
A sound knowledge of TAGs as an energy store with some reasons given. Some knowledge of energy comparisons with other energy sources	2–3
- Basic knowledge of TAGs as an energy store OR - One reason for TAGs being used as an energy store	1

Question 30 (c) (i)

Outcomes assessed: H7, H9, H10, H13

MARKING GUIDELINES

Criteria	Marks
Names correct acid	1

Question 30 (c) (ii)

Outcomes assessed: H2, H4, H8, H13

MARKING GUIDELINES

Criteria	Marks
Demonstrates a thorough knowledge of the characteristics and functions of Type 1 and Type 2 muscles	3–4
Demonstrates a sound knowledge of the characteristics and functions of Type 1 and Type 2 muscles	2
Demonstrates a basic knowledge of the characteristics or functions of Type 1 and Type 2 muscles	1

Question 30 (d) (i)

Outcomes assessed: H2, H3, H8, H9, H11, H13

MARKING GUIDELINES

Criteria	Marks
Demonstrates a thorough knowledge of the use of appropriate experimental procedures to test the influence of one factor on the reaction of a named enzyme	2
Demonstrates limited knowledge of the use of appropriate experimental procedures to test the influence of one factor on the reaction of a named enzyme	1

Question 30 (d) (ii)

Outcomes assessed: H2, H3, H7, H8, H9, H11, H13, H14

MARKING GUIDELINES

Criteria	Marks
Uses correct results to reach a correct conclusion based on an extensive knowledge of enzyme structure	4
- Uses correct results to reach a correct conclusion based on a sound knowledge of enzyme structure OR - Uses correct results to reach a conclusion based on a thorough knowledge of enzyme structure	2–3
- Gives correct results of experiment OR - States a correct conclusion OR - Shows a limited knowledge of enzyme structure	1

Question 30 (e)

Outcomes assessed: H2, H3, H4, H8, H9, H13

MARKING GUIDELINES

Criteria	Marks
Shows an extensive knowledge of the oxidation of fatty acids as an inhibitor of the conversion of pyruvate to acetyl Co A and links this to the dietary needs of sprinting athletes	5–6
Shows a sound knowledge of the oxidation of fatty acids as an inhibitor of the conversion of pyruvate to acetyl Co A	3–4
Shows a limited knowledge of the oxidation of fatty acids as an inhibitor of the conversion of pyruvate to acetyl Co A	1–2

Section II

The Chemistry of Art

Question 31 (a) (i)

Outcomes assessed: H2, H3, H4, H7, H11, H13

MARKING GUIDELINES

Criteria	Marks
• Correctly identifies metal ion	1

Question 31 (a) (ii)

Outcomes assessed: H2, H3, H4, H7, H11, H13

MARKING GUIDELINES

Criteria	Marks
• Provides a thorough explanation of how a flame colour is produced	2–3
• Correctly identifies ONE aspect associated with production of a flame colour	1

Question 31 (b)

Outcomes assessed: H2, H3, H4, H6, H10, H13

MARKING GUIDELINES

Criteria	Marks
• Provides features of bonding in a coordination complex using an example from medicine or biological research	3–4
• Identifies a correct aspect of bonding and identifies a coordination complex OR • Sound description of bonding	2
• Identifies a coordination complex OR • Identifies one correct aspect of bonding	1

Question 31 (c) (i)

Outcomes assessed: H2, H3, H7, H13

MARKING GUIDELINES

Criteria	Marks
Provides a sound explanation of the nature of a reflectance spectrum	1–2

Question 31 (c) (ii)

Outcomes assessed: H2, H3, H4, H7, H13

MARKING GUIDELINES

Criteria	Marks
As for below and must mention that to identify composition of a pigment a comparison with a known standard is needed	3
Describes the use of infrared/ultraviolet reflectance/absorption spectra	1–2

Question 31 (d) (i)

Outcomes assessed: H2, H3, H6, H7, H13

MARKING GUIDELINES

Criteria	Marks
Correctly states the electronic configuration	1

Question 31 (d) (ii)

Outcomes assessed: H2, H6, H11

MARKING GUIDELINES

Criteria	Marks
Demonstrates a thorough knowledge of the use of appropriate experimental procedures in determining the oxidising strength of potassium permanganate	1–2

Question 31 (d) (iii)

Outcomes assessed: H2, H6, H7, H8, H9, H10, H13, H14

MARKING GUIDELINES

Criteria	Marks
Uses correct results to justify conclusion and includes correct half equations	3
- Justifies a conclusion based on results OR - States correct results and includes correct half equations	2
- States correct results OR - Writes a correct half equation OR - Justifies conclusion without reference to the results	1

Question 31 (e)

Outcomes assessed: H2, H6, H13, H14

MARKING GUIDELINES

Criteria	Marks
Demonstrates an extensive knowledge of the physical properties of the elements in the first transition series and draws relationships between these properties	5–6
Provides an analysis of the trend in one physical property of the elements in the first transition series and refers to other properties	3–4
Identifies correctly one trend in one physical property of the elements in the first transition series	2
- Identifies one physical property of the elements in the first transition series OR - Identifies the first transition series	1

Section II

Forensic Chemistry

Question 32 (a) (i)

Outcomes assessed: H2, H9

MARKING GUIDELINES

Criteria	Marks
• Correctly identifies the functional group	1

Question 32 (a) (ii)

Outcomes assessed: H8, H9, H13

MARKING GUIDELINES

Criteria	Marks
• Correctly compares several aspects of the chemical reactions	3
• Compares one or two aspects of the reactions correctly OR • Gives correct details of several aspects of one of the reactions of one compound	2
• Shows a basic knowledge of reactions of the compounds with KMnO_4	1

Question 32 (b)

Outcomes assessed: H1, H3, H4, H12, H13

MARKING GUIDELINES

Criteria	Marks
• Demonstrates a thorough knowledge of the uses of each technique in forensic chemistry (with an example) and discusses the value of each technique	3–4
• Demonstrates a sound knowledge of the processes of each technique OR • Makes a statement of the value of each technique	1–2

Question 32 (c) (i)

Outcomes assessed: H3, H4, H8, H9, H13

MARKING GUIDELINES

Criteria	Marks
• Correctly identifies the class of compound	1

Question 32 (c) (ii)

Outcomes assessed: H1, H3, H4, H13

MARKING GUIDELINES

Criteria	Marks
• Demonstrates an extensive knowledge of how both processes separate organic compounds	4
• Demonstrates a thorough knowledge of how both processes separate organic compounds	3
• Sound knowledge of how both processes separate organic compounds	2
• Basic knowledge of how both processes separate organic compounds OR • Sound knowledge of ONE process	1

Question 32 (d) (i)

Outcomes assessed: H1, H2, H3, H7

MARKING GUIDELINES

Criteria	Marks
• Correctly identifies the piece of equipment	1

Question 32 (d) (ii)

Outcomes assessed: H1, H2, H3, H7, H11, H13

MARKING GUIDELINES

Criteria	Marks
• Demonstrates a thorough knowledge of the procedure used to determine the emission spectrum of sodium	1–2

Question 32 (d) (iii)

Outcomes assessed: H1, H2, H3, H4, H7, H11, H13

MARKING GUIDELINES

Criteria	Marks
• Demonstrates an extensive knowledge of the transition of electrons between different energy levels and applies this to the method used to generate a sodium spectrum in the lab	3
• Demonstrates a sound knowledge of the way emission spectra are formed. Gives only general information	2
• Demonstrates a basic knowledge of emission spectra	1

Question 32 (e)*Outcomes assessed: H1, H3, H4, H5, H13***MARKING GUIDELINES**

Criteria	Marks
• Demonstrates an extensive knowledge of the uses of DNA analysis in forensic chemistry	5–6
• Demonstrates a sound knowledge of the uses of DNA analysis in forensic chemistry	3–4
• Demonstrates a basic knowledge of the uses of DNA analysis in forensic chemistry OR • Gives examples of uses of DNA analysis	1–2